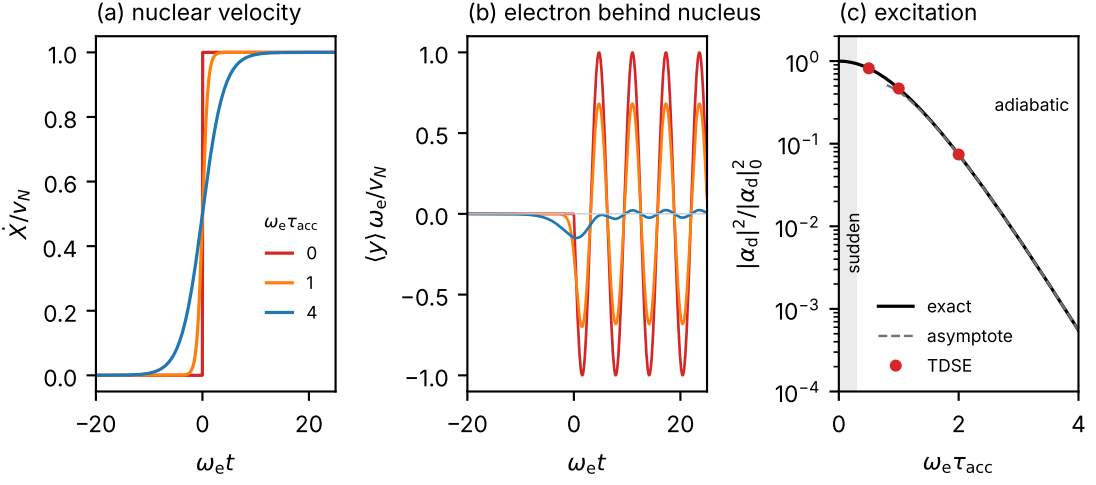
(a) nuclear velocity
$\dot{X}/v_N$
1.0
0.8
0.6
0.4
0.2
0.0
−20
0
20
$\omega_e t$
$\omega_e\tau_{acc}$
0
1
4
(b) electron behind nucleus
$\langle y\rangle\,\omega_e/v_N$
1.0
0.5
0.0
−0.5
−1.0
−20
0
20
$\omega_e t$
(c) excitation
$|\alpha_d|^2/|\alpha_d|^2_0$
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
adiabatic
sudden
exact
asymptote
TDSE
0
2
4
$\omega_e\tau_{acc}$